Dynamic Planet KEY

2022 Golden Gate Science Olympiad Invitational Dynamic Planet Test

By Brian Amaro & Chloe Cheng

Time limit: 50 minutes

Reminders:

- When applicable, numerical answers in a reasonable range will be accepted.
- Show all work for calculations—give us the chance to award partial credit!
- Be concise when asked to explain. Generally, anything over a few sentences might be too much.
- There is no penalty for guessing; answer as much as you can.

Good luck!

Part 0: An Introduction to the Foundations of Contemporary Hydrology

1. (1 pt) What *is* a river?

   Any mention of flowing water will suffice

Part 1: Water We Doing?

2. (1 pt) Scoria is best described to have:

    A. Low porosity; high permeability
    B. High porosity; low permeability
    C. Low porosity; low permeability
    D. High porosity; high permeability

3. (1 pt) What relationship does the amount of canopy coverage have with direct throughfall onto the soil?

    A. A direct relationship
    B. An inverse relationship
    C. No significant relationship
    D. Relationship depends on vegetation type

4. (1 pt) Around which value is the typical pH of normal rain?

    A. 3
    B. 5
    C. 7
    D. 9

5. (1 pt) Which of these tidal wetland types matches the description "Tidal water covers the land surface less often than daily"?

    A. Subtidal
    B. Irregularly exposed
    C. Regularly flooded
    D. Irregularly flooded

6. (2 pts) Select all of the following that the thawing of permafrost would cause.

    A. Increase the permeability of the subsurface
    B. Decrease the permeability of the subsurface
    C. Not affect the permeability of the subsurface
    D. Would cause the size of a small lake to increase
    E. Would cause the size of a small lake to decrease

7. (1 pt) Which of the following is typically the warmest part of a lake?

    A. The supralimnion
    B. The epilimnion
    C. The metalimnion
    D. The hypolimnion

8. (1 pt) Lake turnover occurs because of the weakening of:

    A. Thermocline
    B. Halocline
    C. Epicline
    D. Trade winds

9. (1 pt) A tectonic lake is most likely to form in a:

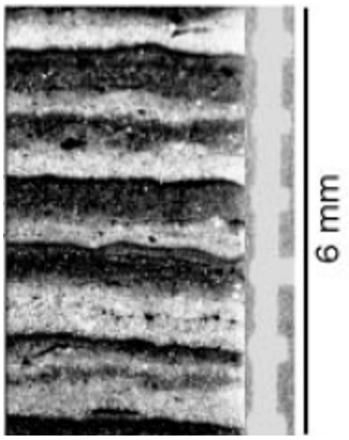
    A. Horst
    B. Graben
    C. Transpression
    D. Transform fault

10. (1 pt) Which of the following is not a characteristic of a bog?

    A. Has a low peat content
    B. Is also known as a muskeg
    C. Has acidic water
    D. Most water received through precipitation

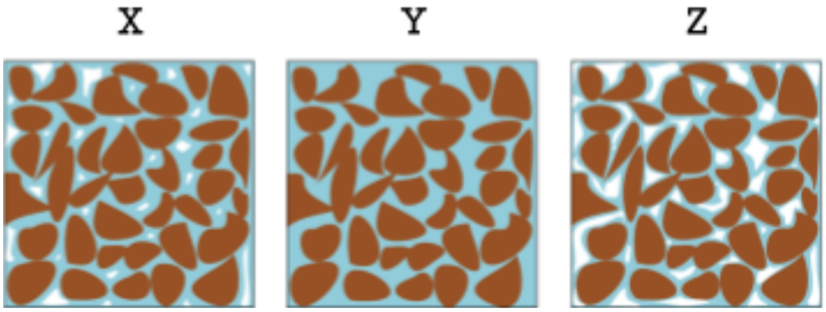
11. (1 pt) Which of the following can support trees?

    A. Bogs
    B. Marshes
    C. Swamps
    D. Fens



12. (2 pts) Choose all of the following that accurately describe the formation of the sedimentary sequence above.

A. **The alternating dark and light bands represent deposition during different seasons.**
B. The alternating dark and light bands represent the different climate states of icehouse Earth and greenhouse Earth.
C. **The white bands form during conditions with a higher energy level than the dark bands.**
D. **This type of sequence is particularly common in lacustrine environments.**
E. This type of sequence is particularly common in fluvial environments.
F. This type of sequence is particularly common in marine environments.



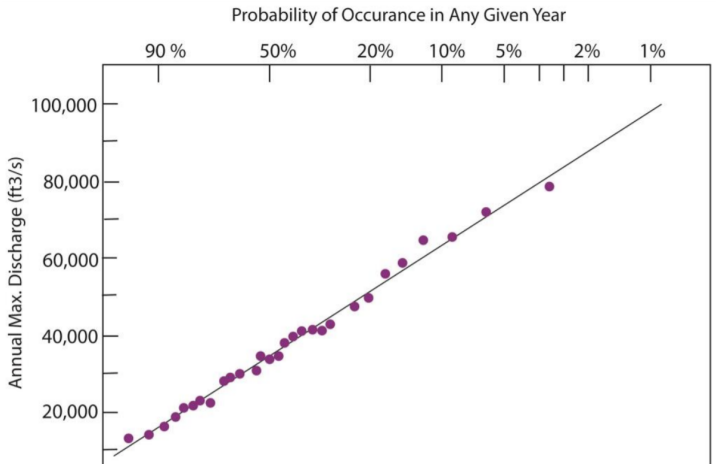
13. (1 pt) **Figure B** shows illustrations of cross sections of soil at different soil moisture thresholds. Which of the following correctly matches the diagram to the threshold?

A. X → Saturation. Y → Field capacity. Z → Permanent Wilting Point.
B. **X → Field capacity. Y → Saturation. Z → Permanent Wilting Point.**
C. X → Saturation. Y → Permanent Wilting Point. Z → Field capacity.
D. X → Field capacity. Y → Permanent Wilting Point. Z → Saturation.
E. X → Permanent Wilting Point. Y → Field capacity. Z → Saturation.
F. X → Permanent Wilting Point. Y → Saturation. Z → Field capacity.

Part A.2 Water You Doing?

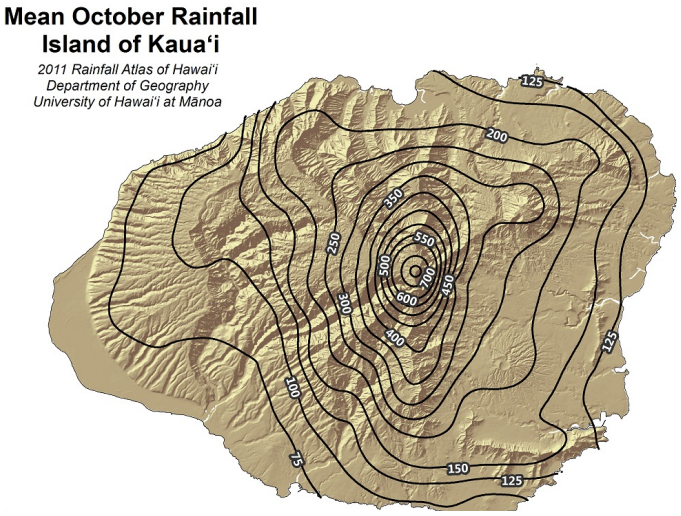
14. (2 pts) Imagine you model a stream as two connected semicircles each with a radius of 15 meters such that the stream forms a "S" shape. What would be the sinuosity of the stream?

    30*π / 60 = π/2 = 1.571



15. (1 pt) Using the flood frequency chart above, what is the probability of a flood with a max discharge of 50,000 ft^3/s occurring in any given year?

    1/5 = .2 = 20%



16. Using the map above, answer the following two questions.

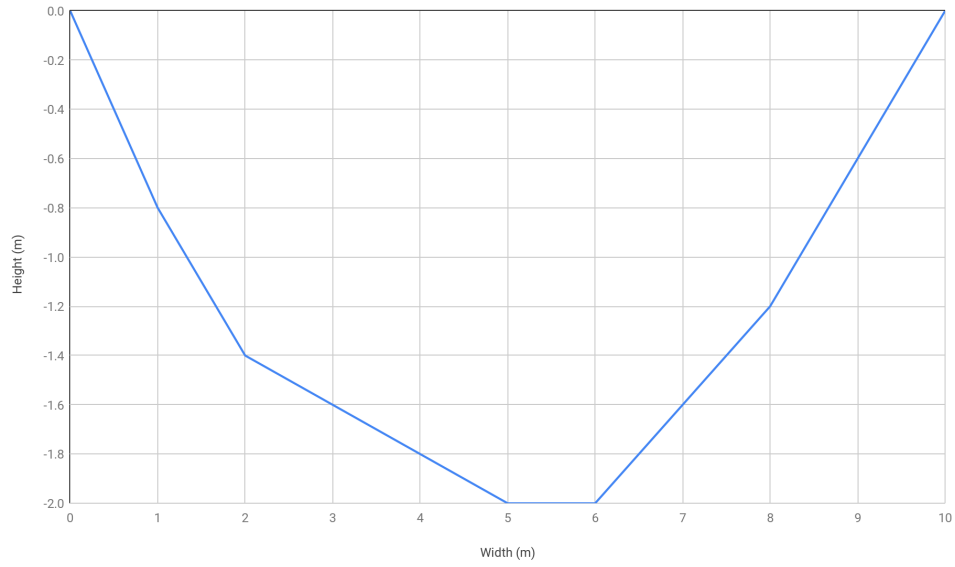
a. (1 pt) What are the names of the black contour lines on the map?

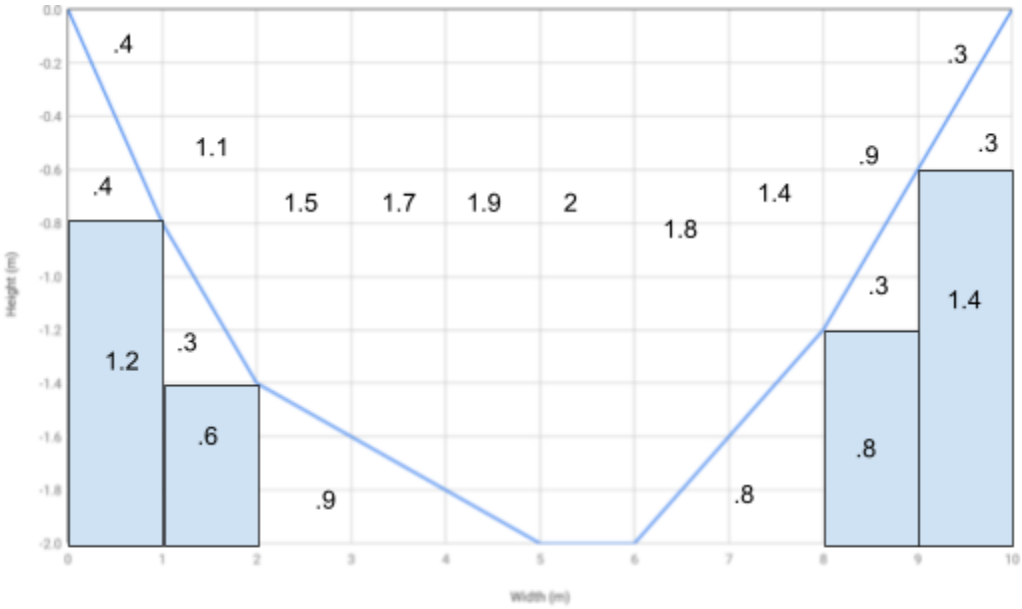
Isohyets

b. (2 pts) Briefly explain why there are higher values closer to the mountain.

Orographic lifting (or more general explanation of the same concept)

(4 pts) Using data from a real stream in Vermont, we have provided a graph of the cross-section of the stream. Using an average velocity of 7 m/s, find the discharge of the stream.





20 total 20 -1.6 -.9 -.9 -1.7 -1.1 -.8= 13

.4 +1.1 +1.5 +1.7 +1.9 +2 +1.8 +1.4 +.9 +.3 = 13

Discharge = Area * X meters cubed / second

See annotated diagram for work, but discharge = 13 * 7 = 91 m^3/s

Grading: 1 pt for attempting to estimate area and using discharge formula If answer in range (91, 91.1), then 3 points; Elif answer in range (86, 96), then 2 points; Elif answer in range (80, 101), then 1 point; Else 0 points.-0.5 pts if no units

Part 2: Ripples Believe It Or Not!



Dried silt. Tiretracks (upper left) and earthworm (lower right) for scale.

17. (1 pt) Indicate the direction of water movement in the image above.

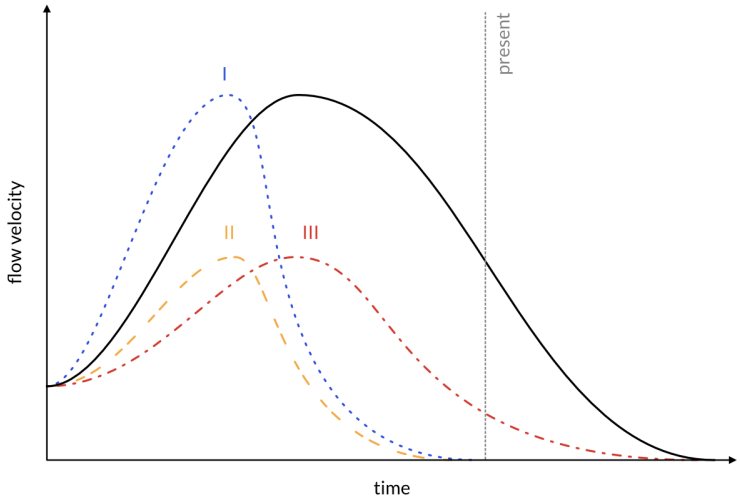
    A. Left to right
    B. Right to left
    C. Both directions



Wet silt on the side of the street after a rainy day. Earthworm track (center) for scale.

18. (1 pt) Indicate the direction of water movement in the image above.

    A. Left to right
    B. Right to left
    C. Both directions



19. (1 pt) The black curve in the graph above depicts the changes in flow velocity over time that produced the ripples in the first image. Assuming all other conditions are the same, which curve most likely produced the features in the foreground of the second image?

    A. I
    B. II
    C. III

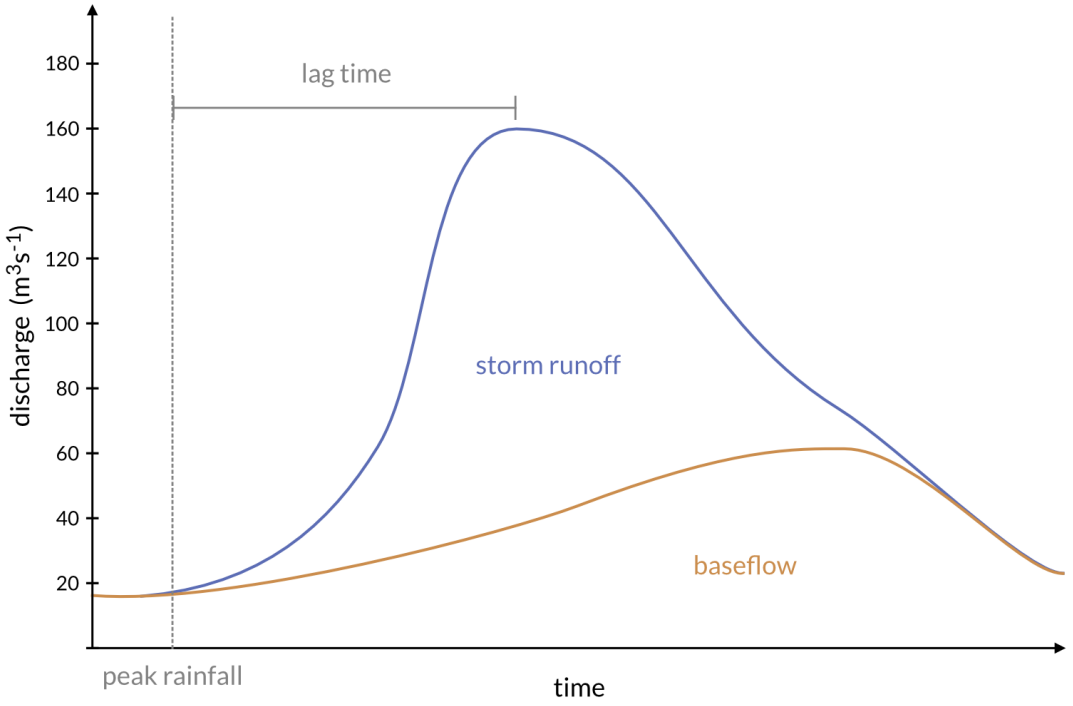
20. (1 pt) Which of the following about the above curves best determines how likely the highest ripple size over a span of time is preserved after water flow ceases?

    A. Initial flow velocity
    B. Slope of rising limb
    C. Height of peak
    D. Slope of falling limb

21. (4 pts) Between the flow structures in the first and second image, which is likely more prevalent in rock records? Explain using your answer from 4) (i.e., how is the chance of preservation related to initial flow velocity, rising limb slope, peak height, or falling limb slope?).

    The flow structures in the first image are more likely to be preserved because larger ripples or dunes require higher flow velocities, and lower flow is more common than higher flow (+1). Additionally, even if larger ripples form at one point, they will likely be "overwritten" as flow velocity gradually decreases (+2). The lower the slope of the falling limb (+1), the more sediments will reorganize into structures corresponding to lower flow

For the remaining questions in this section, refer to the following storm hydrograph. Precipitation begins at the start of the x-axis.



22. (1 pt) Further urban development in the watershed upstream would have what result on the hydrograph for a similar precipitation event?

    A. An increase in lag time
    B. A higher peak flow
    C. Both A and B
    D. Neither A or B

23. (1 pt) How would urbanization most plausibly affect baseflow?

    A. Baseflow would be higher on average, making the hydrograph broader in width.
    B. Baseflow would be higher on average, giving the hydrograph two distinct peaks.
    C. Baseflow would be lower on average, making the hydrograph more symmetrical.
    D. Baseflow would be lower on average, the hydrograph maintaining its overall shape.

24. (1 pt) Which of the following best accounts for the delay from the onset of precipitation to the contribution of storm runoff to stream flow?

    A. Runoff takes time to reach stream gauges.
    B. The ground surface is initially unsaturated.
    C. Peak rainfall is reached after precipitation begins.
    D. Streamflow consists entirely of baseflow.

25. (4 pts) At the location of the stream gauge at the time of peak discharge, the cross sectional area is 85 square meters and the wetted perimeter is 24 m. Calculate the hydraulic gradient at the stream gauge using the Chézy formula, taking the Chézy coefficient to be 20 $m^{1/2}/s$.

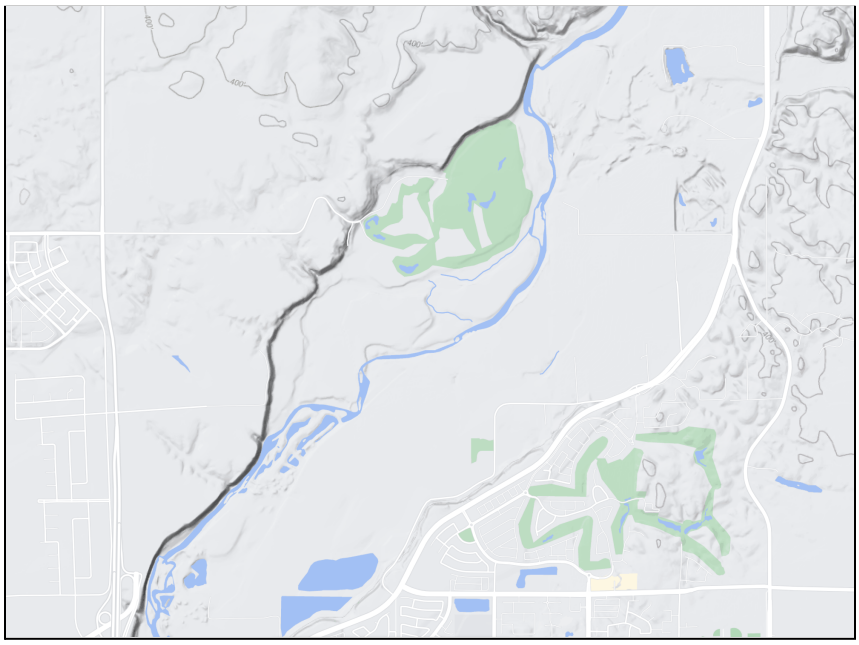
    A. 0.0025 m/m
    B. 18.0706 m/m
    C. 51.6398 m/m
    D. 476.0952 m/m
    E. 0.1245 m/m

    $V = C*sqrt(R*i) = Q/A$

    $i = (Q/AC)^2/R = (160\ m^3/s/((85\ m^2)(20\ m^{1/2}/s)))2/(85\ m^2/24\ m) =$ 0.0025 m/m

Part 3: In Plain Sight, Sort Of

Use the following image for questions 27-30.



Section of the San Joaquin River slightly north of Fresno. Multiple terraces can be seen within the floodplain.

26. (1 pt) Which of the following describes the most likely composition of the terraces as compared to the current floodplain?

    A. No sediment
    B. Similar sediments
    C. Coarser sediment
    D. Finer sediment

27. (1 pt) If we know that the river was a graded river at some point in time, what can be inferred about changes in stream maturity of the San Joaquin River to account for the formation of these terraces?

    A. Only increased
    B. Only decreased
    C. Increased and decreased multiple times
    D. Neither increased or decreased

28. (2 pts) Select all of the following that is/are true when a terrace is forming compared to when a terrace is not forming.

    A. Greater deposition at point bars
    B. Greater erosion at cutbanks
    C. Greater deposition in the river bed
    D. Greater erosion in the river bed

29. (1 pt) Which of the following could have been responsible for creating the terraces?

    A. The over pumping of groundwater
    B. The damming of the river downstream
    C. An increase in sea level
    D. Tectonic uplift in the region

30. (1 pt) Natural levees form on either side of a river with floods. Which of the following best characterizes the distribution of sediment size about natural levees?

    A. Grains become coarser with distance from the river channel on both sides
    B. Grains become finer with distance from the river channel on both sides
    C. Grains become coarser with distance from the river channel inside meanders and finer outside meanders
    D. Grains become finer with distance from the river channel inside meanders and coarser outside meanders

31. (2 pts) Give a justification for your answer to question 31.

    After water tops the levees, it slows with distance and decreases in competence. The flood water is first unable to carry the coarsest sediments, and gradually the finer sediments.

32. (1 pt) Flood stage is the water height at which a river will flood. How is the flood stage of the river above affected with successive floods?

    A. Increases
    B. Decreases
    C. Stays the same
    D. Becomes zero

33. (2 pts) Give a justification for your answer to question 33.

Each successive flood deposits more sediments atop natural levees. With time, a greater height is needed to surpass the taller levees and create a flood.

34. (2 pts) Select all of the following that would promote the formation of floodplain wetlands:

A. Well-drained soils
B. Underlying clay beds
C. Underlying sandy beds
D. Frequent flooding

35. (2 pts) Give one reason why tall levees (natural or artificial) may **reduce** a water source to wetlands.

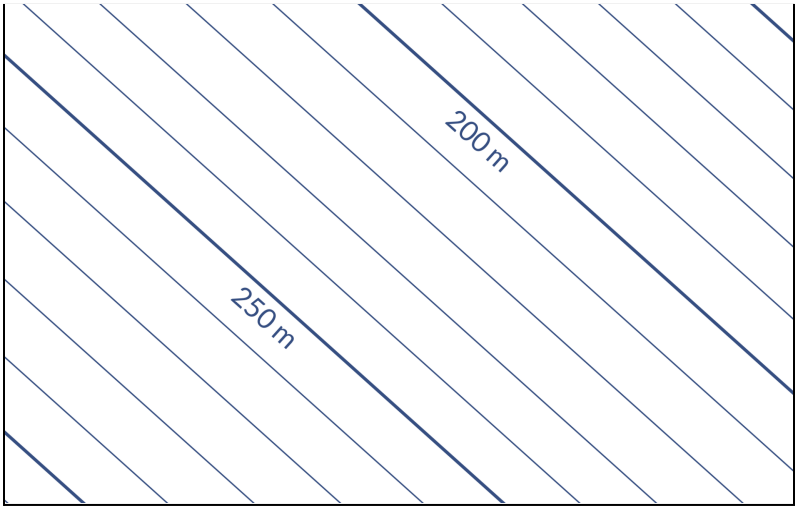
Less water enters the floodplain from flooding; flooding is less frequent.

36. (2 pts) Give one reason why tall levees (natural or artificial) may **increase** a water source to wetlands.

More precipitation will remain in the floodplain rather than flowing directly into the river as runoff.

Part 4: London Underground but Not Necessarily London

Consider this set of simplified water table contours for questions 38-42.



37. (1 pt) In what direction would groundwater flow?

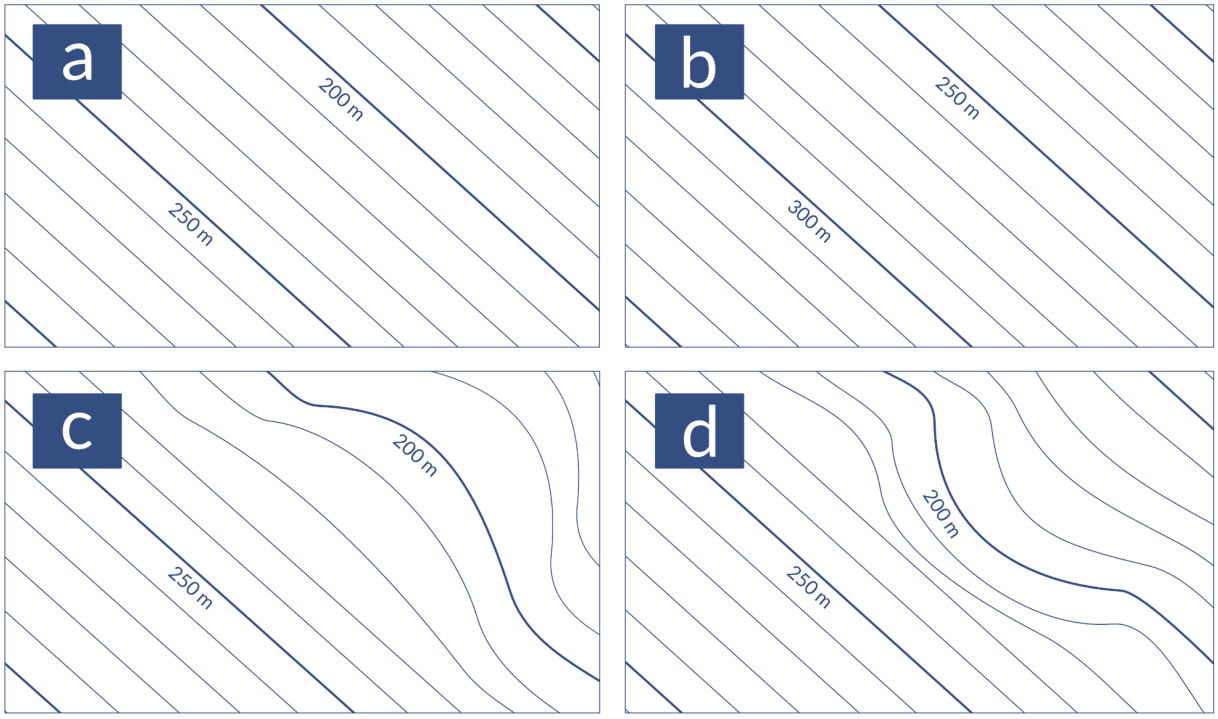
    A. **Bottom left to top right**
    B. Top right to bottom left
    C. Bottom right to top left
    D. Top left to bottom right

38. (1 pt) Assuming everything unspecified remains the same, which of the following would increase hydraulic gradient the most?

    A. **Halving the distance between contour lines**
    B. Doubling the distance between contour lines
    C. Subtracting 200 m from each contour
    D. Adding 200 m to each contour

39. (2 pts) What additional piece(s) of information is (are) necessary to calculate the hydraulic gradient between any two given points on the map above?

    A. Direction of groundwater flow
    B. **The horizontal scale of the image**
    C. Height intervals of the subcontours
    D. Topography of the overlying land surface



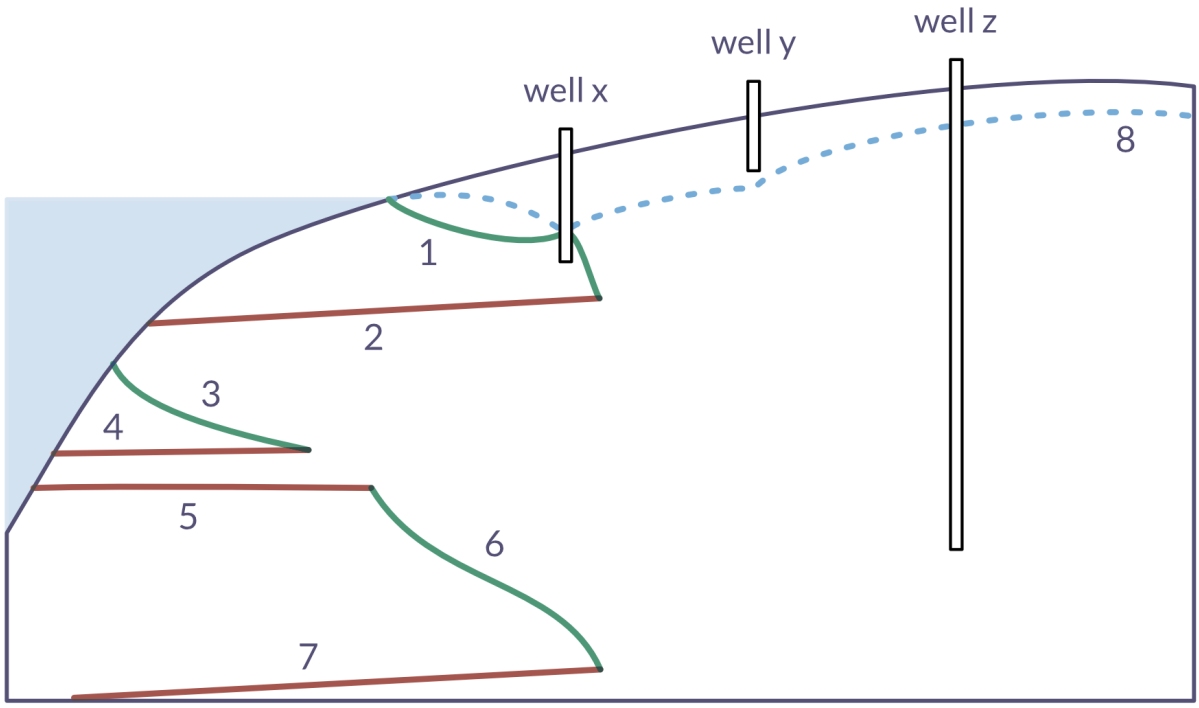
40. (1 pt) Suppose a recharge basin is constructed directly above the 200 m water table contour. What is a plausible contour map given significant recharge?

A
B
C
D

41. (1 pt) Which letter from the previous question would result from extensive groundwater pumping?

A
B
C
D

For the remaining questions in the section, refer to the diagram below.



Land-ocean interface with a cross section of aquifers that extend across the figure. Select boundaries are labeled lines 1-8 and wells are labeled x, y, and z. Assume all features referenced are located within the same plane.

List all lines 1-8 that apply to the statement. Lines may be used more than once or not at all.

42. (1.5 pt) An aquifer boundary.

    2, 4, 5, 7

43. (1.5 pt) Exists within a confined aquifer.

    3, 6

44. (1.5 pt) Restricts upward water movement.

    5

45. (1.5 pt) Contacts an unsaturated zone.

    8

46. (1.5 pt) An interface between freshwater and saltwater.

    1, 3, 6

47. (1.5 pt) Changes position with human water use.

    1, 3, 6, 8

48. (4 pts) A drill core was taken to the depth of well Z just before it was constructed in order to study the subsurface. A portion of the core appears to have a coarse-grained material overlying a very fine-grained material.

    a. What line(s) could this core have come from?

       2, 4, 7

    b. Justify your response by describing the two materials in the context of groundwater.

       These are the lower boundaries of aquifers. Aquifers in this case are made up of the relatively coarse-grained material observed in the core. Meanwhile, aquicludes are made up of the fine-grained material.

49. (2 pts) Well Z was drilled deep after well Y went dry. Select all of the following that is/are true about well Z:

    A. It caused line 1 to move to the right of its original position.
    B. It caused line 3 to move to the right of its original position.
    C. **It caused line 6 to move to the right of its original position.**
    D. It will eventually form a depression in line 8.
    E. It will eventually form a depression in line 5.

50. Pumpage well X was stopped after the crops it irrigated began to dehydrate.

    a. (1 pt) What general term is used for the phenomenon of line 1 moving landwards?

       Saltwater intrusion

    b. (2 pts) What prevents it from continuing to move landwards?

       Groundwater flow towards the ocean

c. (1 pt) Well Y no longer pumps water because it was within the ___ of well X.

Cone of depression

51. (1 pt) The figure shows that a stream flowing on the surface would likely lose water to:

A. Evaporation
B. Evapotranspiration
C. Runoff
D. Infiltration

52. (3 pts) The stream mentioned previously was contaminated by an industrial plant near well Y. Which of lines 1-8 marks the deepest contaminants can reach, and why?

Contaminants would not travel below line 2, which represents the bottom of the aquifer into which contaminants percolated through from the stream.

53. (3 pts) Describe how a scenario in which the height of line 8 is changed would allow contaminants to remain at the surface.

If line 8 was higher and at the stream level, groundwater would flow into the stream, thus preventing contaminants from traveling out of the stream.

54. Sea level rises and falls over long time periods with influences from tectonics, climate, and more.

a. (2 pts) Suppose sea level falls to a height between lines 5 and 7. Nearest to which two lines would we expect to find springs?

Lines 2 and 4

b. (2 pts) Of the water produced by those two springs, which probably has spent more time as groundwater? Justify your answer.

Water from the spring at line 4 has probably spent more time as groundwater as part of the lower aquifer beneath an impermeable bed. Groundwater in the upper aquifer has a shorter residence time.

Part 5: Lake Me Up For Lake's Sake

55. (1 pt) Lakes can be created by a variety of basin-forming processes. Crater Lake, a lake atop Mount Mazama in the Cascade Range and site of the famous Wizard Island, is best classified as:

    A. **Volcanic**
    B. Tectonic
    C. Fluvial
    D. Glacial

56. (3 pts) Considering your answer to the previous question, explain why water loss to groundwater is limited in Crater Lake.

    The basin Crater Lake lies in is made up of igneous rock, which is generally impermeable.

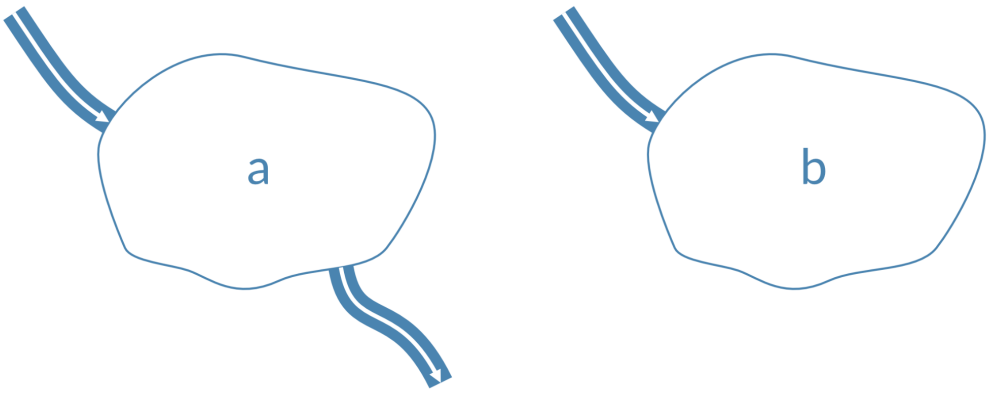
57. (2 pts) What is the drainage pattern that feeds into Crater Lake termed?

    Centripetal

58. (2 pts) What drainage pattern drains the remainder of runoff on Mount Mazama?

    Radial

For questions 60-61, refer to the figure below.



Lake A has an inlet and an outlet while Lake B only has an inlet. The lakes and their inlets are identical otherwise.

59. Differentiate water loss from lakes A and B by naming the primary reason for water loss from each lake.

    a. (1 pt) Lake A:

    Flow through the outlet

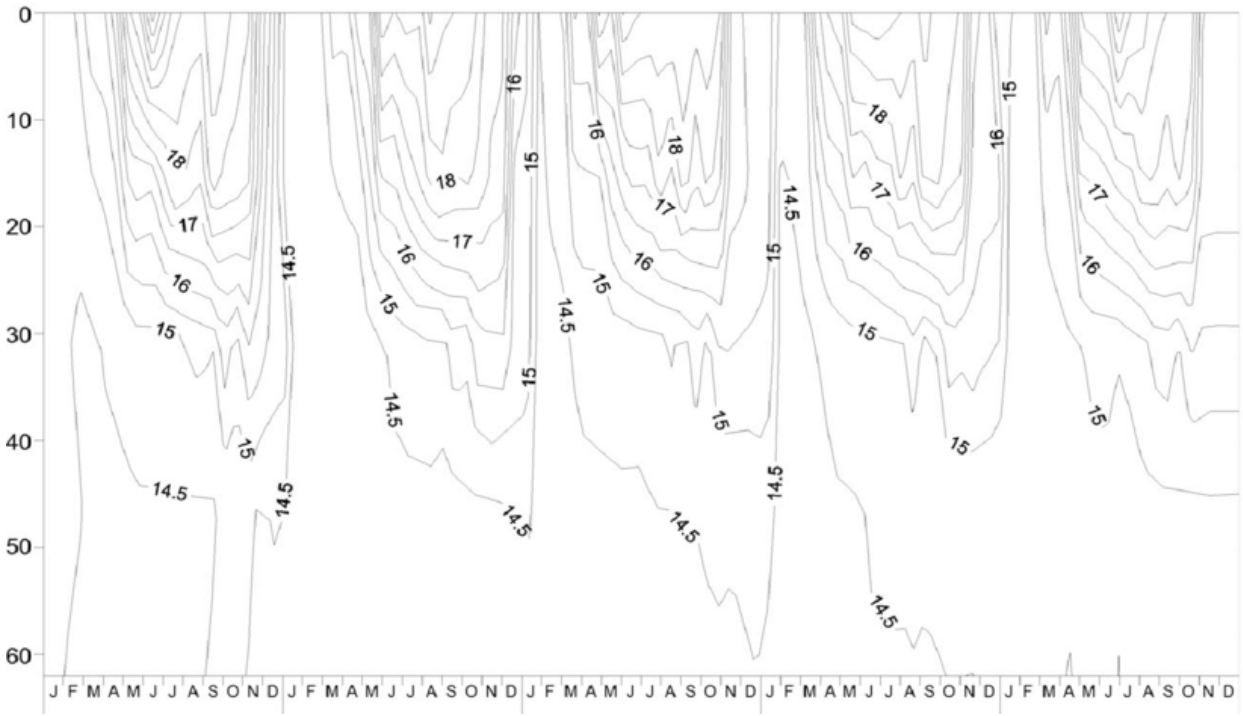
    b. (1 pt) Lake B:

    Evaporation (infiltration also accepted)

60. (1 pt) How do the contents of Lake B compare to Lake A, assuming the stream loads of their inlets are the same?

    A. Greater ratio of sodium to potassium ions
    B. Greater ratio of phosphorus to nitrogen
    C. Higher concentration of dissolved load
    D. Greater amount of sands and gravels

For questions 62-64, refer to the following depth-time diagram of isotherms in degrees Celsius for a lake in the Northern Hemisphere (credit).



61. (1 pt) The isotherms in show that:

A. The warmest average lake temperature occurs around December.
B. **Temperature varies with depth the most in the summer.**
C. Water temperature at the lake bottom varies more than at the surface.
D. Surface temperature changes horizontally in uniform distance intervals.

62. (1 pt) In terms of stratification patterns, this lake is best described as:

A. Monomictic
B. Dimictic
C. Polymictic
D. Amictic

63.

a. (2 pt) Approximately when do waters in this lake vertically mix the most?

A. Spring
B. Summer
C. Autumn
D. Winter

b. (3 pts) Explain how you could determine whether the lake is closer to the poles or equator using your answer to 9.a.

The lake lacks thermal stratification in the winter when the warmer epilimnion cools enough to match the temperature of the cooler hypolimnion, which is characteristic of warm monomictic lakes typically located at lower latitudes.

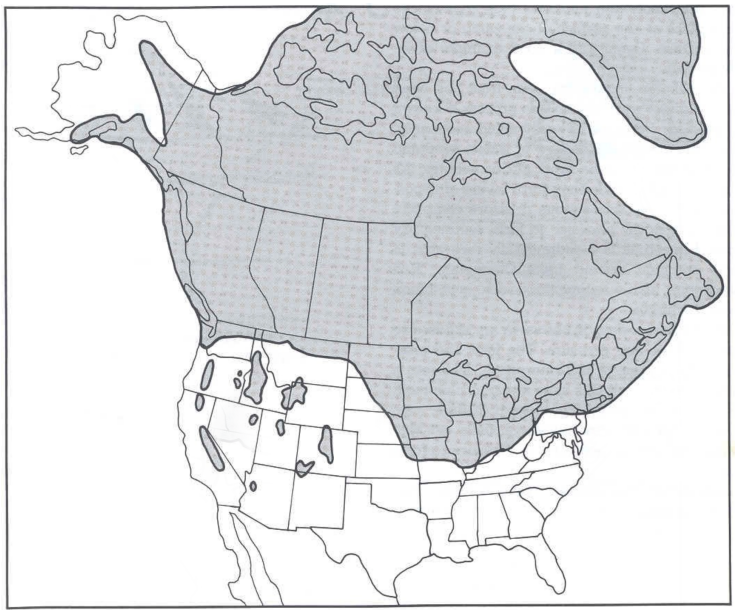
Part 6: More than 3%: Hydrology and Climate

64. (1 pt) Limestone formations can indicate past shallow marine environments, which are exposed to the surface with a lower relative sea level. Groundwater is able to permeate limestone easily because:

    A. Calcium carbonate is highly soluble in pure water.
    B. Rainwater typically contains dissolved carbon dioxide.
    C. Limestone is much less dense than other rocks.
    D. Circulating groundwater is warm near the surface.

65. (2 pts) Sinkholes can form from the collapse of overlying material in karst landscapes. Briefly explain how pumping groundwater can cause sinkholes to suddenly form.

    Lowering the water table can expose existing voids in limestone, removing the hydrostatic pressure that supported the roof above it and allowing it to collapse.



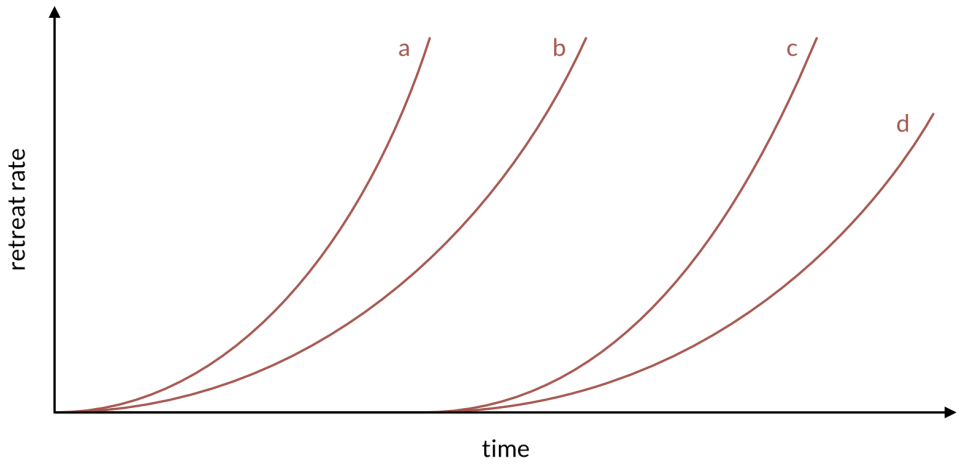
66. The shaded area in the figure above contains permeable rocks formed in the last couple tens of thousands of years.

    a. (1 pt) Identify what the shaded area represents.

    The shaded area represents the extent of glaciation in North America.

b. (3 pts) In a couple of sentences, explain how climate might have influenced the formation of aquifers in this scenario.

A colder climate can produce ice sheets. With eventual warming, glaciers deposit permeable ground moraines as they retreat, leading to the widespread formation of aquifers.



Graph comparing four different scenarios of glacial retreat rate over time, where the leftmost time is the time when the glacier reaches its maximum extent.

67. Suppose a proglacial lake formed at the terminus of a retreating glacier, where water is dammed by a terminal moraine.

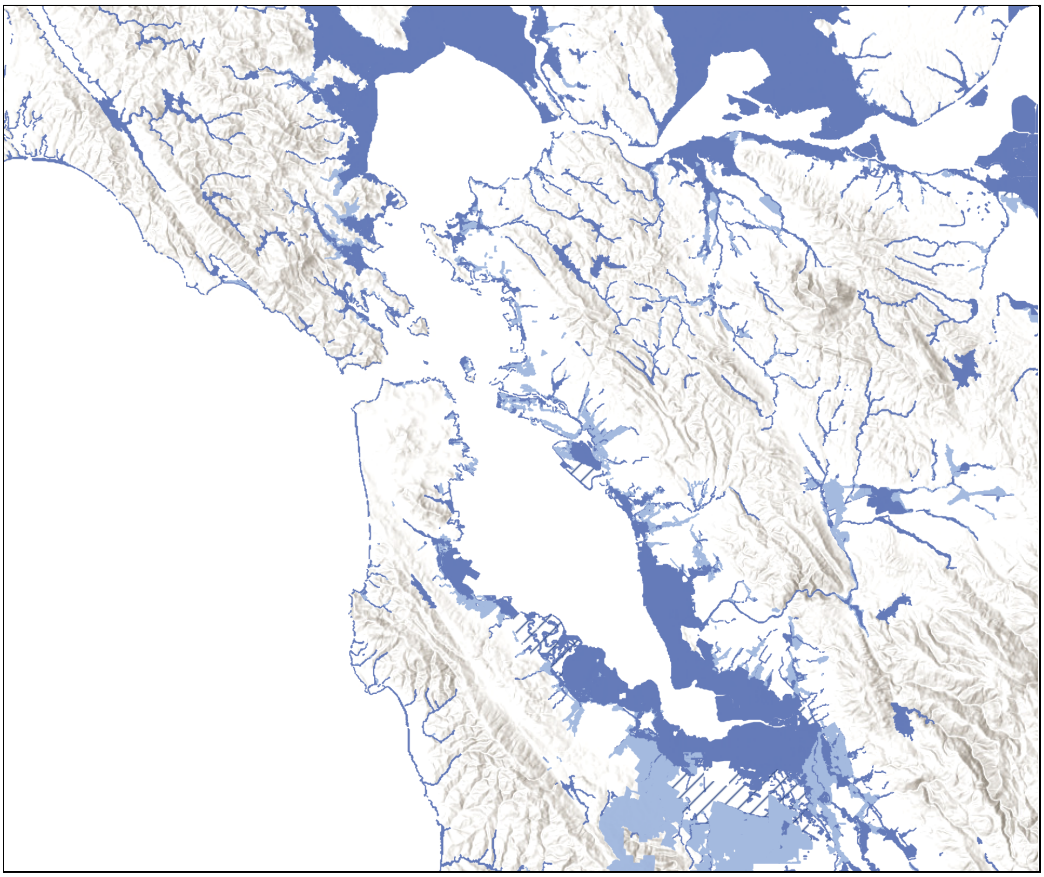
    a. (3 pts) How would the height of terminal moraines in scenarios a and b compare to scenarios c and d? Justify your answer.

    Terminal moraines from A and B would be lower than those in C and D. C and D form higher moraines because the glacier terminus spends more time in one location, depositing more sediment.

    b. (1 pt) Which scenario would likely produce the deepest proglacial lake?

    A
    B
    C
    D

For the remaining questions in the section, refer to the flood map below.



Federal Emergency Management Agency (FEMA) flood map of the Bay Area. Dark blue indicates areas inundated by a 100-year flood.

68. (1 pt) Flood maps are typically used to assess and mitigate flood risk. Which of the following most accurately describes every point within the dark blue areas in the map?

A. Flooded once every 100 years
B. Flooded at least once every 100 years
C. Has a 1% chance of flooding in one year
D. Has at least a 1% chance of flooding in one year

69. (3 pts) Diagonal lines in the map show areas that are not considered to be in the flood zone because they are protected by artificial levees. Explain why flooding may actually pose more of a risk to low-elevation homes in these protected areas than higher-elevation homes inside the flood zone.

    Even if the risk of levees failing is low, floods that are large enough to top the levees at all could cause catastrophic damage in submerging low-lying buildings. The same flood would certainly flood the higher-elevation buildings as well, but the damage may be much less severe.

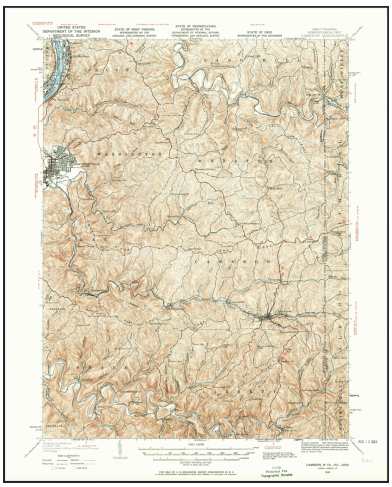
70. (3 pts) Identify **two** causes of Bay Area flooding by storms, one over land and one over the ocean.

    Heavy precipitation, storm surge.

71. (3 pts) Climate change is rapidly changing the nature of severe weather events, including those that cause floods. Give one reason why a 10-year flood a century ago may not be equivalent to a 10-year flood now.

    Climate change tends to increase the intensity, duration, and/or frequency of flood-causing storms. The strength of the current 10-year flood is likely greater than the past one. Floods with the same strength as the past 10-year flood likely occur more often than an average of every 10 years, giving it an annual chance greater than 10%.

Part 7: Where's Wa(ter)ldo?



For the following questions,

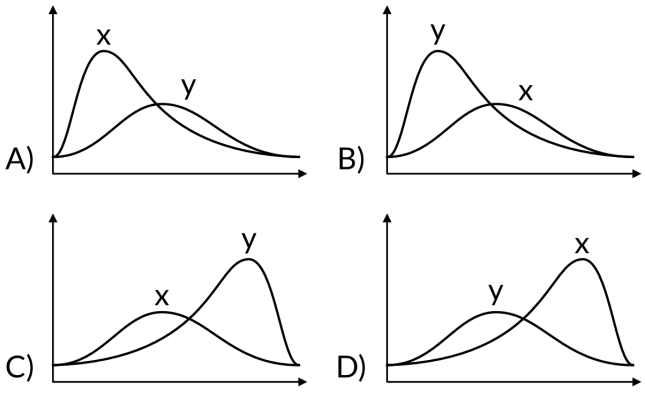
- click here for the labeled map
- click here for a full resolution, unlabeled version

72. Find the streams marked "★" and "☆."

a. (2 pts) In a few words, describe the shapes of the drainage basins drained by ★ and ☆.

★ : elongated, ☆ : rounded

b. (1 pt) If the discharge at ★ and ☆ was measured after a storm, how would the two hydrographs compare?



A
B
C
D

73. (1 pt) What direction is the river in the top left of the map flowing?

A. North
B. East
C. South
D. West

74. (1 pt) As denoted by stickers, which location is most likely to become an oxbow lake after time?

A. W
B. X
C. Y
D. Z

75. (1 pt) Imagine someone living where Toms Run meets Middle Grave Creek sees plastic debris floating along. Which city would be most likely to blame for this pollution?

A. McMehan
B. Moundsville
C. Knoxville
D. Andersonville

End (with a surprise)

